



Algorithm Design

Project

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Your project contains 2 parts:

First, Image compression:

Abstract:

Compression is used about everywhere. Images are very important documents nowadays; to work with them in some applications they need to be compressed, more or less depending on the purpose of the application. The need for an efficient technique for compression of Images ever increasing because the raw images need large amounts of disk space seems to be a big disadvantage during transmission & storage. There are various algorithms that performs this compression in different ways; some are lossless and keep the same information as the original image, some others loss information when compressing the image. In this paper we proposed the Lossless method of image compression and decompression using a simple coding technique called Huffman coding. This technique is simple in implementation and utilizes less memory.

Introduction:

Compression refers to reducing the quantity of data used to represent a file, image or video content without excessively reducing the quality of the original data. Image compression is the application of data compression on digital images. There are different techniques and they all have their own advantages and disadvantages.

Fundamental for compression:

Compression can be divided into two categories, as Lossy and Lossless compression.

Lossy compression means that some data is lost when it is decompressed. Lossy compression bases on the assumption that the current data files save more information than human beings can "perceive". Thus, the irrelevant data can be removed.

Lossless compression means that when the data is decompressed, the result is a bit-for-bit perfect match with the original one. The name lossless means "no data is lost", the data is only saved more efficiently in its compressed state, but nothing of it is removed.

Compression Principle:

In an image there are three types of redundancies in order to compress file size. They are:

- a) **Coding redundancy:** Fewer bits to represent frequently occurring symbols.
- b) **Interpixel redundancy:** Neighboring pixels have almost same value.
- c) **Psycho visual redundancy:** Human visual system cannot simultaneously distinguish all colors.

Compression Ratio (CR):

Compression ratio (CR) is defined as number of bits to represent the size of original image to the number of bits to represent the size of compressed image. Compression ratio shows that how much time the image has been compressed.

Implementation:

Based on the descriptions above, you should implement an image compression that is lossless. One way to do that is using Huffman Coding algorithm which is based on the frequency of occurrence of a data item i.e., pixel in images. The technique is to use a lower number of bits to encode the data in to binary codes that occurs more frequently.

Your program should satisfy these needs:

- Read image
- Encode it to Huffman code.
- Generate Huffman code tree and save it in a text file.
- Decode the generated Huffman code into the image.
- Calculate the CR.

Second, Image steganography:

Abstract:

Steganography is the study and practice of concealing information within objects in such a way that it deceives the viewer as if there is no information hidden within the object. Simply put, it is hiding information in plain sight, such that only the intended recipient would get to see it. In order to satisfy this, need you should prepper a secret key so that only the people with key can crack the code!

Types of steganography techniques:

There are several types and forms of steganography, and in the sections below, we'll explore a few interesting and commonly used ones. Broadly steganography techniques could be classified into

- Physical: One that does not involve the use of digital mediums or files. Examples of such techniques include
 1. Passing messages written with invisible ink, which can then be read by the intended recipient by applying certain chemical techniques
 2. By using ciphering techniques to hide information within textual information. Here is an example of a null cypher technique used by a prisoner to convey a secret message to his outfit, which was intercepted and decoded by the FBI. See graphic below, which looks like an ordinary letter, however, when one looks at every fifth word (highlighted in red) it reveals a plot to murder someone i.e, refer to the last few red words "IF GUILTY OF WRITEUP, HE SHOULD BE HIT."
- Microdots involve shrinking messages to such tiny dimensions, they are made almost invisible. These also involve positioning 'hard-to-see' dots within a message to convey a specific message.
- Digital: Involves the usage of digital mediums such as hiding information within
 1. Text files
 2. Image or picture files
 3. Audio files
 4. Video files

Image Steganography algorithms:

There are several types of image steganography algorithms, but we recommend these two:

1. LSB algorithm
2. PVD algorithm

But you are free to use other algorithms too.

Implementation:

After you compressed image now it's the time to hide something into it! By use of your preferred algorithm, you should now hide a message in the image from last part, hide your message into it and then provide an environment for receiver to find out the message.

Your program should satisfy these needs:

- Read image from last part
- Hide message into it with use of your preferred algorithm
- Receiver can understand the code in a specific way where no other can

Also, you should hand out a short paper about your used algorithm. Format of the paper will be given to you.

- Upload the expected files in a zipped file to the page of the course on Quera.
- Do NOT copy from your classmates' answers :)
- Feel free to ask any questions.

Good Luck!

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